Multimedia Systems Mini Project

Project Specification Sheet • OMC AI Section

Submission: GitHub + Demo Video via Google Form

1. Project Overview

This project introduces you to **multimedia AI systems**. You will build **OCR (image → text)** and **ASR (audio → text)** pipelines, then combine them into one unified system.

Task	What You Build
OCR — Optical Character Recognition	Extract text from images
ASR — Automatic Speech Recognition	Convert speech / audio into text

You will complete **three levels of increasing complexity** — beginner-friendly at Level 0, with optional challenges for advanced students.

Starter code is provided on GitHub: https://github.com/b-fatma/checkpoint.1.4

It includes basic OCR and ASR scripts with sample images and audio to get you started.

By the end of this project, your system should:

- Read text from an image
- Transcribe speech from audio
- Optionally, combine both outputs into one final text document, and summarize the notes and generate an audio summary
- Be built entirely with **open-source tools and pretrained models**

2. Project Levels (0 -> 3)

Level 0 — Starter (Beginner)

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0- Choose a fun name for your project.

1- Set up pytesseract for OCR and OpenAI Whisper for ASR using the starter code repo.

2- Run the provided starter code on sample images and audio.

3- Run it on your own images and audio, and include metrics:
    • ASR: Word Error Rate (WER)
    • OCR: Character or word accuracy (simple ratio of correct characters/words)
    • Observations: Which type of images/audio worked best? Which failed?

4- Try another open-source library for OCR or ASR (e.g., EasyOCR, Vosk) and compare performance.
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Goal: Understand baseline OCR + ASR performance and get familiar with the starter scripts.

Level 1 — Intermediate (Improvement & Analysis)

- OCR: Apply image preprocessing using OpenCV or Pillow:
  - Grayscale, thresholding, denoising, deskewing, resizing
  - Compare text extraction accuracy with metrics
- ASR: Apply optional audio preprocessing:
  - Noise reduction, trimming silence, resampling
  - Compare transcription accuracy
- Bonus: Add a Text-to-Speech (TTS) component:
  - Convert your combined text output to audio (e.g., using Coqui TTS)
  - Demonstrate a read-aloud version of your notes

Goal: Improve system quality and experiment with preprocessing.

Level 2 — Advanced (Multimodal System)

Build a fully multimodal note-taking system:

- Read text from images (OCR)
- Transcribe speech from audio (ASR)
- Combine outputs into a single document
- Optionally summarize the text using a large language model (LLM)
- Optionally convert summary to audio with TTS

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Example Theme: Smart Classroom Assistant

Scan lecture slides + record audio from lectures → produce combined notes
Generate a short summary and audio recap
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Alternative Theme Ideas:

Theme	Use Case
Smart Classroom Assistant	Reads slides + transcribes lectures
Multimodal Note-Taker	Scans handwritten notes + voice memos
Accessibility Tool	Reads text aloud from images + captions speech
Meeting Assistant	Scans agenda + transcribes discussion
Study Assistant	Extracts textbook text + records verbal summaries
Lecture Transcriber	Captures whiteboard content + live speech
Document Scanner with Voice Notes	Digitises paper docs + attaches voice annotations

Goal: Integrate OCR + ASR + optional LLM + TTS into a useful real-world system

3. Suggested Tools

Task	Suggested Tools
OCR	Tesseract • EasyOCR • TrOCR (Hugging Face)
ASR	OpenAI Whisper • wav2vec2 • Vosk
Image Processing	OpenCV • Pillow • scikit-image
General	Python 3 • Hugging Face Transformers • NumPy
LLM	Hugging Face Transformers, OpenAI API (optional for summary)
TTS	Coqui TTS, pyttsx3

4. Deliverables

#	Deliverable	Details
1	**GitHub Repository**	OCR script, ASR script, combine script, sample data, output .txt files, README
2	**Demo Video (2–4 min)**	Google Drive or YouTube (Unlisted). Must show both systems, explain preprocessing, explain how OCR & ASR work, and discuss failure cases.
3	**Google Form Submission**	Name, project title, GitHub link, demo video link

6. Grading Rubric (100 points)

Category	Points
OCR Working	20
OCR Preprocessing	10
ASR Working	20
Combined Output	15
Code Organisation	10
README	10
Demo Video	25 ⭐
TOTAL	**100**

Don't forget to have fun! 🚀
Good luck and enjoy the build.

OMC AI Section Leadership
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